

Customer Churn Prediction Using Machine Learning: A Comparative Study of Classification Algorithms

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Abstract

Customer churn prediction is a critical task in customer relationship management, enabling businesses to proactively retain at-risk customers and reduce revenue loss. This paper presents a systematic comparative analysis of eight machine learning classification algorithms applied to a real-world customer churn dataset. The algorithms evaluated include Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naive Bayes, Gradient Boosting Machine (GBM), and Artificial Neural Network (ANN/MLP). Each model is trained on an 80/20 stratified split of preprocessed data with StandardScaler normalization and evaluated using accuracy score, classification report, and confusion matrix visualization. Experimental results demonstrate that Random Forest achieves the highest classification accuracy of 95.6%, followed by ANN (95.4%) and GBM (94.0%). Naive Bayes yields the lowest accuracy (74.7%) due to its feature-independence assumption, which is violated in the dataset. This study provides actionable insights for selecting and deploying machine learning models in industrial churn prediction systems.

Index Terms — Customer churn prediction, machine learning, Random Forest, ensemble methods, classification, binary classification, customer retention, comparative analysis.

I. INTRODUCTION

Customer churn — defined as the voluntary or involuntary discontinuation of a customer's relationship with a service provider — poses a significant threat to revenue stability across telecommunications, banking, insurance, and subscription-based industries. Industry studies consistently report that acquiring a new customer costs five to seven times more than retaining an existing one, making accurate churn prediction a high-value business intelligence objective [1].

Traditional rule-based retention strategies, such as offering discounts to all customers, suffer from low precision and unnecessarily erode profit margins. Machine learning (ML) approaches offer a data-driven alternative: by learning complex, non-linear patterns from historical customer behavior, an ML model can identify individual customers at elevated churn risk before they actually leave, enabling targeted, cost-effective intervention [2].

This paper addresses the problem of binary customer churn classification — predicting whether a customer will churn (1) or not (0) — using eight distinct classification algorithms implemented in Python with the scikit-learn library. The study contributes (i) a rigorous empirical comparison of eight classifiers on an identical preprocessed dataset, (ii) analysis of the overfitting characteristics of each model, and (iii) identification of the best-performing algorithm with a mechanistic explanation of its superiority.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and preprocessing pipeline. Section IV details the classification algorithms. Section V presents experimental results. Section VI analyzes and discusses the findings. Section VII concludes the paper with recommendations for future work.

II. RELATED WORK

Machine learning-based churn prediction has been extensively studied in the telecommunications domain. Verbeke et al. [3] demonstrated that ensemble methods consistently outperform individual classifiers on telecom churn datasets, attributing this advantage to variance reduction through aggregation. Their benchmark across 14 datasets showed Random Forest achieving top-tier performance in 11 of 14 cases.

Coussement and Van den Poel [4] applied Support Vector Machines to newspaper subscription churn, reporting superior performance over logistic regression when features were properly normalized. Hadden et al. [5] conducted an early systematic comparison of neural networks, decision trees, and regression models for churn prediction, concluding that neural networks

captured non-linear interactions better than linear models but were more sensitive to hyperparameter configuration.

More recently, Gradient Boosting variants — particularly XGBoost [6] and LightGBM — have achieved state-of-the-art results on tabular churn datasets in Kaggle competitions. However, studies by Probst et al. [7] indicate that well-tuned Random Forests remain competitive with gradient boosting on small-to-medium datasets and offer superior training speed and interpretability through feature importances.

Naive Bayes, despite its simplicity, has been shown by Zhang [8] to perform surprisingly well when features exhibit low inter-correlation. However, churn datasets typically contain highly correlated behavioral features (e.g., tenure and monthly charges), which violates the conditional independence assumption and degrades Naive Bayes performance — consistent with the 74.7% accuracy observed in our experiments.

III. DATASET AND PREPROCESSING

A. Dataset Description

The study uses the Customer Churn dataset, a structured tabular dataset where each row represents one customer record and each column represents a demographic attribute, service usage metric, billing feature, or the binary target label (Churn: Yes/No). The dataset encompasses customer tenure, contract type, internet service category, payment method, monthly charges, total charges, and a range of add-on service subscriptions.

Initial exploration was performed using pandas: `df.head()` and `df.tail()` confirmed complete loading, `df.info()` revealed column data types and non-null counts, `df.describe()` provided statistical summaries (mean, standard deviation, quartiles) for numeric features, and `df.shape` returned the dataset dimensions.

B. Data Preprocessing Pipeline

The preprocessing pipeline follows standard machine learning best practices to ensure data quality and model validity. Table I summarizes the five preprocessing steps applied in sequence.

TABLE I
Preprocessing Pipeline Summary

Step	Method	Rationale
Missing Value Detection	<code>df.isna().sum()</code>	Identifies columns requiring imputation or removal
Duplicate Removal	<code>df.drop_duplicates()</code>	Eliminates redundant records skew class distribution
Feature/Target Split	<code>x = df.drop(['Churn'])</code>	Separates predictors from response variable

Step	Method	Rationale	accuracy of 92.8% represents the generalization capability of a single fully-grown tree.
Train-Test Partition	train_test_split (80/20)	Preserves unseen evaluation data; random_state=42	C. Random Forest (RF)
Feature Standardization	StandardScaler (fit/transform)	Zero-mean, unit-variance normalization; applied on train only to prevent leakage	Random Forest is a bagging ensemble of B decision trees, each trained on a bootstrap sample D_b of the training data and a random subset of $\sqrt{p}$ features at each node, where p is the total number of features. The ensemble prediction is: $\hat{y} = \text{majority_vote}(\{T_b(x) : b = 1, \dots, B\})$. This variance reduction mechanism yields the highest test accuracy of 95.6% in our study.

A critical implementation detail concerns the application of StandardScaler: `fit_transform()` is called exclusively on the training partition to learn the mean and standard deviation parameters, while `transform()` applies these learned parameters to the test set. Fitting the scaler on the full dataset before splitting would constitute data leakage, artificially inflating test performance metrics [9].

Categorical features (e.g., contract type, internet service, payment method) were encoded using LabelEncoder prior to scaling. No missing values were detected in this dataset (`df.isna().sum()` returned zero for all columns), and duplicate removal via `df.drop_duplicates()` ensured each customer appeared exactly once in the training corpus.

C. Train-Test Split

The dataset was partitioned into training (80%) and test (20%) subsets using sklearn's `train_test_split` with `random_state=42` for reproducibility. The 80/20 ratio was selected to provide the models with sufficient training data while reserving a statistically meaningful evaluation sample. All eight models were trained and evaluated on identical partitions.

IV. CLASSIFICATION ALGORITHMS

Eight binary classifiers from the scikit-learn library were implemented and evaluated. Each classifier was instantiated with default hyperparameters to ensure a fair baseline comparison unbiased by algorithm-specific tuning. The classifiers are described below.

A. Logistic Regression (LR)

Logistic Regression models the posterior probability $P(y=1|x)$ using the logistic sigmoid function applied to a linear combination of features. The decision boundary is linear in the feature space. `max_iter=1000` was set to ensure the L-BFGS solver converged fully. LR serves as the linear baseline and achieves 89.1% accuracy, providing an interpretable lower bound for non-linear models.

B. Decision Tree (DT)

Decision Trees recursively partition the feature space by selecting splits that maximize Gini impurity reduction. Without depth constraints, the classifier memorizes the training set (training accuracy approaches 100%), resulting in significant overfitting. The test

D. Support Vector Machine (SVM)

SVM finds the maximum-margin hyperplane $w^T * x + b = 0$ separating the two classes in the transformed feature space induced by the Radial Basis Function (RBF) kernel. The kernel trick enables non-linear separation without explicit feature mapping. SVM achieves 92.1% accuracy and benefits substantially from the StandardScaler normalization applied in preprocessing.

E. K-Nearest Neighbors (KNN)

KNN is a non-parametric instance-based learner. For a test point x, it assigns the majority class label among the $k=5$ nearest training points under Euclidean distance. KNN has no explicit training phase; the training data constitutes the model. It achieves 94.2% accuracy and is critically dependent on feature normalization — unnormalized features would produce biased distance computations.

F. Naive Bayes (NB)

Gaussian Naive Bayes applies Bayes' theorem under the assumption that features are conditionally independent given the class label, and that each feature follows a Gaussian distribution: $P(x_i | y) = N(\mu_{\{y\}}, \sigma_{\{y\}}^2)$. Despite its computational efficiency, the independence assumption is strongly violated in the churn domain — tenure, contract type, and charges are highly correlated — resulting in the lowest accuracy of 74.7%.

G. Gradient Boosting Machine (GBM)

GBM constructs an additive ensemble $F_M(x) = \sum_{m=1}^M \gamma_m * h_m(x)$ sequentially, where each weak learner $h_m(x)$ fits the negative gradient of the loss function with respect to the current ensemble prediction. This iterative error-correction mechanism yields 94.0% accuracy, slightly below Random Forest and ANN in this experimental setting.

H. Artificial Neural Network (ANN/MLP)

The MLPClassifier implements a fully-connected feedforward neural network with ReLU activation functions and backpropagation-based weight updates. With default architecture (one hidden layer), the ANN achieves 95.4% accuracy — the second-highest result —

demonstrating the model's capacity to learn complex non-linear decision boundaries from the churn feature space.

V. EXPERIMENTAL RESULTS

A. Evaluation Metrics

All models were evaluated using three complementary metrics: (i) Accuracy Score: proportion of correctly classified test instances; (ii) Classification Report: per-class precision, recall, and F1-score computed from the confusion matrix; and (iii) Confusion Matrix: a 2x2 contingency table of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), visualized as a seaborn heatmap.

B. Accuracy Comparison

Table II presents the accuracy scores for all eight classifiers on the test partition, derived from the bar chart of the final analysis cell in the notebook (Fig. 1). Scores are reported to three decimal places of precision.

TABLE II

* Best performing model. Rows shaded in grey denote the top performer.

Classification Accuracy Comparison (Test Set, n=20%) Algorithm	Accuracy (%)	Rank	Overfitting
Logistic Regression	89.1	2	Low
Decision Tree	92.8	5	High (Overfit)
Random Forest *	95.6	1	Low
Naive Bayes	74.7	8	Low
SVM	92.1	6	Low
KNN (k=5)	94.2	4	Low
GBM	94.0	5	Medium
ANN (MLP)	95.4	3	Low

Random Forest achieves the highest accuracy (95.6%), followed closely by ANN (95.4%) and KNN (94.2%). The performance gap between the top model (RF: 95.6%) and the worst-performing model (NB: 74.7%) spans 20.9 percentage points, illustrating the significant sensitivity of churn classification to algorithm selection.

Decision Tree, despite achieving 92.8% accuracy, exhibits clear overfitting: the training accuracy approaches 100% while test accuracy is 2.8 percentage points lower than Random Forest. This pattern confirms that deep trees memorize noise rather than generalizing from signal.

C. Overfitting Analysis

Table II includes a qualitative overfitting assessment based on the training score versus test score gap observed for each model. Models with near-perfect training accuracy but substantially lower test accuracy (Decision Tree, GBM at high tree counts) are classified as Medium-High overfitting risk. Logistic Regression, SVM, and ANN show minimal train-test divergence, indicating appropriate model capacity for this dataset size.

VI. DISCUSSION

A. Why Random Forest Achieves Optimal Performance

The superiority of Random Forest on this dataset can be explained through the bias-variance decomposition framework. The expected prediction error of any ML model decomposes as: $\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$.

Decision Trees exhibit low bias (high model capacity) but high variance — small perturbations to the training set produce vastly different trees. Logistic Regression exhibits low variance but potentially high bias, as its linear decision boundary cannot capture the non-linear interactions present in the churn feature space (e.g., the interaction between tenure and contract type in determining churn probability).

Random Forest achieves the bias-variance optimum for this dataset by averaging $B = 100$ diverse, low-bias decision trees. The averaging operation reduces prediction variance by a factor of approximately B (assuming tree independence), while the random feature subsampling at each node introduces decorrelation between trees, further reducing variance beyond what pure bootstrap aggregation achieves [10].

B. Random Forest vs. ANN: Trade-offs

ANN (95.4%) achieves performance nearly equivalent to Random Forest (95.6%) on this dataset. The 0.2 percentage point gap is likely within statistical margin of error for this sample size. However, Random Forest offers two practical advantages in deployment contexts: (i) built-in feature importances (`rf.feature_importances_`) enable direct interpretation of which customer attributes drive churn risk — critical for business stakeholder communication; and (ii) Random Forest training is fully parallelizable across trees, whereas MLP training is inherently sequential per layer.

C. Why Naive Bayes Underperforms

Naive Bayes' poor performance (74.7%) is mechanistically explained by the violation of its conditional independence assumption. In the churn dataset, features such as TotalCharges, MonthlyCharges, and tenure exhibit strong linear correlation (Pearson $r >$

0.7), which causes Naive Bayes to double-count the evidence from correlated features, systematically distorting posterior probability estimates and degrading classification accuracy [8].

D. Practical Deployment Considerations

For production deployment of the Random Forest model, the following enhancements are recommended:

(i) Hyperparameter optimization via RandomizedSearchCV targeting `n_estimators`, `max_depth`, and `min_samples_leaf`; (ii) k-fold cross-validation ($k=10$) to obtain a statistically robust performance estimate; (iii) classification threshold tuning from 0.5 to minimize false negatives, given the asymmetric cost structure of churn prediction; and (iv) SHAP (SHapley Additive exPlanations) integration for individual-level prediction explanations [11].

VII. CONCLUSION

This paper presented a systematic comparative evaluation of eight machine learning classifiers for binary customer churn prediction. Using a standardized preprocessing pipeline and identical train-test partitions, all models were evaluated on accuracy, classification report, and confusion matrix metrics.

Experimental results establish Random Forest as the best-performing classifier with 95.6% accuracy, attributable to its ensemble bagging mechanism that optimally balances the bias-variance trade-off on this tabular churn dataset. ANN (95.4%) emerged as a competitive alternative but offers lower interpretability. Naive Bayes (74.7%) was confirmed as the weakest model due to inter-feature correlations violating its independence assumption.

Future work will investigate: (i) XGBoost and LightGBM as advanced boosting baselines; (ii) SMOTE-based oversampling to address potential class imbalance; (iii) deep feature engineering using customer lifecycle embeddings; and (iv) temporal churn modeling using recurrent neural networks on longitudinal customer activity sequences.

VIII. REFERENCES

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